PROBLEM1:

```
import java.io.*;


class Student {
    int rollNo;
    String name;
    int marks1, marks2, marks3;

    void acceptDetails(BufferedReader br) throws IOException {
        System.out.print("Enter Roll No: ");
        rollNo = Integer.parseInt(br.readLine());

        System.out.print("Enter Name: ");
        name = br.readLine();

        System.out.print("Enter Marks 1: ");
        marks1 = Integer.parseInt(br.readLine());

        System.out.print("Enter Marks 2: ");
        marks2 = Integer.parseInt(br.readLine());

        System.out.print("Enter Marks 3: ");
        marks3 = Integer.parseInt(br.readLine());
    }
```

```
int calculateTotal() {
    return marks1 + marks2 + marks3;
}

double calculatePercentage() {
    return calculateTotal() / 3.0;
}

String calculateGrade() {
    double percentage = calculatePercentage();

    if (percentage >= 90)
        return "A+";
    else if (percentage >= 80)
        return "A";
    else if (percentage >= 70)
        return "B";
    else if (percentage >= 60)
        return "C";
    else if (percentage >= 50)
        return "D";
    else
        return "F";
}
```

```java
    void displayResult() {
        System.out.println("\n Student Result");
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Total Marks: " + calculateTotal());
        System.out.println("Percentage: " + calculatePercentage() + "%");
        System.out.println("Grade: " + calculateGrade());
    }
}

public class prblem1 {
    public static void main(String[] args) throws IOException {

        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

        Student s1 = new Student();
        Student s2 = new Student();

        System.out.println("Enter details of Student 1:");
        s1.acceptDetails(br);

        System.out.println("\nEnter details of Student 2:");
        s2.acceptDetails(br);
```

```
        s1.displayResult();

        s2.displayResult();

    }

}
```

PROBLEM2:

```
import java.io.*;


class BankAccount {

    private int accountNumber;

    private String accountHolderName;

    private double balance;




    BankAccount() {

        accountNumber = 0;

        accountHolderName = "Unknown";

        balance = 0;

    }




    BankAccount(int accountNumber, String accountHolderName, double balance) {

        this.accountNumber = accountNumber;

        this.accountHolderName = accountHolderName;

        this.balance = balance;

    }




    void deposit(double amount) {

        balance = balance + amount;
```

```
        System.out.println("Deposited: " + amount);
    }

        void withdraw(double amount) {
        if (amount <= balance) {
            balance = balance - amount;
            System.out.println("Withdrawn: " + amount);
        } else {
            System.out.println("Insufficient Balance!");
        }
    }

        void displayAccount() {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Account Holder: " + accountHolderName);
        System.out.println("Balance: " + balance);
    }
}

public class prblem2 {
    public static void main(String[] args) throws IOException {

        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
```

```
    BankAccount account1 = new BankAccount();

System.out.println("Enter details for Account 1:");

System.out.print("Account Number: ");
int number = Integer.parseInt(br.readLine());

System.out.print("Account Holder Name: ");
String name = br.readLine();

System.out.print("Initial Balance: ");
double balance = Double.parseDouble(br.readLine());



BankAccount account2 = new BankAccount(number, name, balance);

System.out.println("\n--- Account Details ---");
account2.displayAccount();

System.out.print("\nEnter deposit amount: ");
double depositAmount = Double.parseDouble(br.readLine());
account2.deposit(depositAmount);

System.out.print("Enter withdrawal amount: ");
```

```
        double withdrawAmount = Double.parseDouble(br.readLine());
        account2.withdraw(withdrawAmount);


        System.out.println("\n--- Updated Account Details ---");
        account2.displayAccount();
    }
}
```

PROBELM3:

```
import java.io.*;


abstract class Shape {

    abstract void calculateArea();

    abstract void calculatePerimeter();

}


class Circle extends Shape {

    double r;


    Circle(double r) {

        this.r = r;

    }


    void calculateArea() {

        double area = Math.PI * r * r;

        System.out.println("Circle Area = " + area);

    }


    void calculatePerimeter() {

        double perimeter = 2 * Math.PI * r;

        System.out.println("Circle Perimeter = " + perimeter);

    }

}
```

```
class Rectangle extends Shape {
    double length, breadth;

    Rectangle(double length, double breadth) {
        this.length = length;
        this.breadth = breadth;
    }

    void calculateArea() {
        double area = length * breadth;
        System.out.println("Rectangle Area = " + area);
    }

    void calculatePerimeter() {
        double perimeter = 2 * (length + breadth);
        System.out.println("Rectangle Perimeter = " + perimeter);
    }
}

class Triangle extends Shape {
    double base, height, side1, side2, side3;

    Triangle(double base, double height, double side1, double side2, double side3) {
```

```
        this.base = base;
        this.height = height;
        this.side1 = side1;
        this.side2 = side2;
        this.side3 = side3;
    }

    void calculateArea() {
        double area = 0.5 * base * height;
        System.out.println("Triangle Area = " + area);
    }

    void calculatePerimeter() {
        double perimeter = side1 + side2 + side3;
        System.out.println("Triangle Perimeter = " + perimeter);
    }
}

public class prblem3 {
    public static void main(String[] args) throws IOException {

        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
```

```
System.out.print("Enter radius of Circle: ");

double r = Double.parseDouble(br.readLine());



System.out.print("Enter length of Rectangle: ");

double length = Double.parseDouble(br.readLine());

System.out.print("Enter breadth of Rectangle: ");

double breadth = Double.parseDouble(br.readLine());



System.out.print("Enter base of Triangle: ");

double base = Double.parseDouble(br.readLine());

System.out.print("Enter height of Triangle: ");

double height = Double.parseDouble(br.readLine());

System.out.print("Enter side 1 of Triangle: ");

double side1 = Double.parseDouble(br.readLine());

System.out.print("Enter side 2 of Triangle: ");

double side2 = Double.parseDouble(br.readLine());

System.out.print("Enter side 3 of Triangle: ");
```

```java
        double side3 = Double.parseDouble(br.readLine());


        Shape s[] = new Shape[3];

        s[0] = new Circle(r);
        s[1] = new Rectangle(length, breadth);
        s[2] = new Triangle(base, height, side1, side2, side3);


        System.out.println("\n--- Shape Details ---");

        for (int i = 0; i < 3; i++) {
            s[i].calculateArea();
            s[i].calculatePerimeter();
            System.out.println();
        }
    }
}
```

PROBLEM4:

```
import java.io.*;

abstract class Employee {
    int employeeId;
    String employeeName;
    double basicSalary;

    Employee(int employeeId, String employeeName, double basicSalary) {
        this.employeeId = employeeId;
        this.employeeName = employeeName;
        this.basicSalary = basicSalary;
    }

    abstract double calculateSalary();

    void display() {
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Employee Name: " + employeeName);
        System.out.println("Basic Salary: " + basicSalary);
    }
}

class PermanentEmployee extends Employee {
```

```
    PermanentEmployee(int id, String name, double salary) {
        super(id, name, salary);
    }

    double calculateSalary() {
        double hra = 0.20 * basicSalary;
        double da = 0.40 * basicSalary;
        double pf = 0.12 * basicSalary;

        double grossSalary = basicSalary + hra + da;
        double netSalary = grossSalary - pf;

        return netSalary;
    }
}

class ContractEmployee extends Employee {

    ContractEmployee(int id, String name, double salary) {
        super(id, name, salary);
    }

    double calculateSalary() {
        double allowance = 0.10 * basicSalary;
```

```
        double grossSalary = basicSalary + allowance;

        return grossSalary;
    }
}

public class prblem4 {
    public static void main(String[] args) throws Exception {

        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

        System.out.print("Enter Employee ID: ");
        int id = Integer.parseInt(br.readLine());

        System.out.print("Enter Employee Name: ");
        String name = br.readLine();

        System.out.print("Enter Basic Salary: ");
        double salary = Double.parseDouble(br.readLine());

        System.out.print("Enter Employee Type (1-Permanent, 2-Contract): ");
        int type = Integer.parseInt(br.readLine());

        Employee e;
```

```
        if (type == 1) {
            e = new PermanentEmployee(id, name, salary);
        } else {
            e = new ContractEmployee(id, name, salary);
        }

        System.out.println("\n Employee Details ");
        e.display();

        System.out.println("Calculated Salary: " + e.calculateSalary());
    }
}
```

PROBLEM5:

```
import java.io.*;


interface Payment {

    void makePayment(double amount);

    void paymentDetails();

}


class CreditCardPayment implements Payment {

    String cardNumber;

    String cardHolderName;

    double amount;


    CreditCardPayment(String cardNumber, String cardHolderName) {

        this.cardNumber = cardNumber;

        this.cardHolderName = cardHolderName;

    }


    public void makePayment(double amount) {

        this.amount = amount;

        System.out.println("\nPayment Successful!");

    }


    public void paymentDetails() {

        System.out.println("Payment Mode : Credit Card");
```

```java
        System.out.println("Card Number : " + cardNumber);
        System.out.println("Card Holder Name : " + cardHolderName);
        System.out.println("Amount : Rs. " + amount);
    }
}

class UPIPayment implements Payment {
    String upiId;
    String userName;
    double amount;

    UPIPayment(String upiId, String userName) {
        this.upiId = upiId;
        this.userName = userName;
    }

    public void makePayment(double amount) {
        this.amount = amount;
        System.out.println("\nPayment Successful!");
    }

    public void paymentDetails() {
        System.out.println("Payment Mode : UPI");
        System.out.println("UPI ID : " + upiId);
```

```java
        System.out.println("User Name : " + userName);
        System.out.println("Amount : Rs. " + amount);
    }
}

class CashPayment implements Payment {
    String customerName;
    double amount;

    CashPayment(String customerName) {
        this.customerName = customerName;
    }

    public void makePayment(double amount) {
        this.amount = amount;
        System.out.println("\nPayment Successful!");
    }

    public void paymentDetails() {
        System.out.println("Payment Mode : Cash");
        System.out.println("Customer Name : " + customerName);
        System.out.println("Amount : Rs. " + amount);
    }
}
```

```
public class prblem5 {
    public static void main(String[] args) throws Exception {

        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

        System.out.println(" PAYMENT SYSTEM ");
        System.out.println("1. Credit Card");
        System.out.println("2. UPI");
        System.out.println("3. Cash");

        System.out.print("Enter Choice: ");
        int choice = Integer.parseInt(br.readLine());

        Payment payment;
        double amount;

        switch (choice) {

            case 1:
                System.out.print("Enter Card Number: ");
                String cardNumber = br.readLine();

                System.out.print("Enter Card Holder Name: ");
```

```
    String cardHolderName = br.readLine();

    System.out.print("Enter Amount: ");
    amount = Double.parseDouble(br.readLine());

    payment = new CreditCardPayment(cardNumber, cardHolderName);
    payment.makePayment(amount);
    payment.paymentDetails();
    break;

case 2:
    System.out.print("Enter UPI ID: ");
    String upiId = br.readLine();

    System.out.print("Enter User Name: ");
    String userName = br.readLine();

    System.out.print("Enter Amount: ");
    amount = Double.parseDouble(br.readLine());

    payment = new UPIPayment(upiId, userName);
    payment.makePayment(amount);
    payment.paymentDetails();
    break;
```

```java
            case 3:
                System.out.print("Enter Customer Name: ");
                String customerName = br.readLine();

                System.out.print("Enter Amount: ");
                amount = Double.parseDouble(br.readLine());

                payment = new CashPayment(customerName);
                payment.makePayment(amount);
                payment.paymentDetails();
                break;

            default:
                System.out.println("Invalid Choice!");
        }

        br.close();
    }
}
```